Assignment 1 R Programming

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```
Diabetes_binary_data = read.csv("~/Health Informatics Program/Fall2026/COMP4033/R Files/diabetes_binary_
```

```
# Install tidyverse package
library("tidyverse")
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
#Print structure of dataset (tidyverse version of str command)
glimpse(Diabetes_binary_data)
```

```
## Rows: 70,692
## Columns: 22
## $ Diabetes_binary      <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ HighBP               <dbl> 1, 1, 0, 1, 0, 0, 0, 0, 0, 0, 1, 0, 1, 1, 0, 0, 0~
## $ HighChol             <dbl> 0, 1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 0, 1, 0, 1, 0, 0~
## $ CholCheck            <dbl> 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ BMI                  <dbl> 26, 26, 26, 28, 29, 18, 26, 31, 32, 27, 24, 21, 2~
## $ Smoker               <dbl> 0, 1, 0, 1, 1, 0, 1, 1, 0, 1, 1, 0, 0, 0, 1, 1, 0~
## $ Stroke               <dbl> 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ HeartDiseaseorAttack <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0~
## $ PhysActivity         <dbl> 1, 0, 1, 1, 1, 1, 1, 0, 1, 0, 1, 1, 1, 0, 1, 1, 1~
## $ Fruits               <dbl> 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0~
## $ Veggies              <dbl> 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 0, 0, 1~
## $ HvyAlcoholConsump    <dbl> 0, 0, 0, 0, 0, 0, 1, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ AnyHealthcare        <dbl> 1, 1, 1, 1, 1, 0, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 1~
## $ NoDocbcCost          <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0~
## $ GenHlth              <dbl> 3, 3, 1, 3, 2, 2, 1, 4, 3, 3, 3, 1, 2, 3, 1, 3, 2~
## $ MentHlth             <dbl> 5, 0, 0, 0, 0, 7, 0, 0, 0, 0, 0, 0, 0, 3, 0, 0, 0~
## $ PhysHlth             <dbl> 30, 0, 10, 3, 0, 0, 0, 0, 0, 6, 4, 0, 0, 3, 0, 0,~
## $ DiffWalk             <dbl> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 1, 0, 0~
## $ Sex                  <dbl> 1, 1, 1, 1, 0, 0, 1, 1, 0, 1, 0, 1, 1, 1, 0, 0, 0~
```

```
## $ Age                  <dbl> 4, 12, 13, 11, 8, 1, 13, 6, 3, 6, 12, 4, 7, 10, 1~
## $ Education            <dbl> 6, 6, 6, 6, 5, 4, 5, 4, 6, 4, 4, 6, 6, 4, 5, 4, 5~
## $ Income               <dbl> 8, 8, 8, 8, 8, 7, 6, 3, 8, 4, 6, 8, 8, 6, 1, 6, 7~
```

```r
# List the variables of the data set/column names
names(Diabetes_binary_data)
```

```
##  [1] "Diabetes_binary"      "HighBP"               "HighChol"
##  [4] "CholCheck"            "BMI"                  "Smoker"
##  [7] "Stroke"               "HeartDiseaseorAttack" "PhysActivity"
## [10] "Fruits"               "Veggies"              "HvyAlcoholConsump"
## [13] "AnyHealthcare"        "NoDocbcCost"          "GenHlth"
## [16] "MentHlth"             "PhysHlth"             "DiffWalk"
## [19] "Sex"                  "Age"                  "Education"
## [22] "Income"
```

```r
# Print the top 15 rows of data set
head(Diabetes_binary_data, n=15)
```

```
##    Diabetes_binary HighBP HighChol CholCheck BMI Smoker Stroke
## 1                0      1        0         1  26      0      0
## 2                0      1        1         1  26      1      1
## 3                0      0        0         1  26      0      0
## 4                0      1        1         1  28      1      0
## 5                0      0        0         1  29      1      0
## 6                0      0        0         1  18      0      0
## 7                0      0        1         1  26      1      0
## 8                0      0        0         1  31      1      0
## 9                0      0        0         1  32      0      0
## 10               0      0        0         1  27      1      0
## 11               0      1        1         1  24      1      0
## 12               0      0        0         1  21      0      0
## 13               0      1        1         1  27      0      0
## 14               0      1        0         1  58      0      0
## 15               0      0        1         1  29      1      0
##    HeartDiseaseorAttack PhysActivity Fruits Veggies HvyAlcoholConsump
## 1                     0            1      0       1                 0
## 2                     0            0      1       0                 0
## 3                     0            1      1       1                 0
## 4                     0            1      1       1                 0
## 5                     0            1      1       1                 0
## 6                     0            1      1       1                 0
## 7                     0            1      1       1                 1
## 8                     0            0      1       1                 0
## 9                     0            1      1       1                 0
## 10                    0            0      1       1                 0
## 11                    1            1      1       1                 0
## 12                    0            1      1       1                 0
## 13                    0            1      1       1                 0
## 14                    0            0      1       1                 0
## 15                    0            1      1       0                 0
##    AnyHealthcare NoDocbcCost GenHlth MentHlth PhysHlth DiffWalk Sex Age
## 1              1           0       3        5       30        0   1   4
```

```
## 2              1          0       3       0        0        0   1  12
## 3              1          0       1       0       10        0   1  13
## 4              1          0       3       0        3        0   1  11
## 5              1          0       2       0        0        0   0   8
## 6              0          0       2       7        0        0   0   1
## 7              1          0       1       0        0        0   1  13
## 8              1          0       4       0        0        0   1   6
## 9              1          0       3       0        0        0   0   3
## 10             1          0       3       0        6        0   1   6
## 11             1          0       3       0        4        0   0  12
## 12             1          0       1       0        0        0   1   4
## 13             1          0       2       0        0        0   1   7
## 14             1          0       3       3        3        0   1  10
## 15             1          0       1       0        0        1   0  10
##    Education Income
## 1          6      8
## 2          6      8
## 3          6      8
## 4          6      8
## 5          5      8
## 6          4      7
## 7          5      6
## 8          4      3
## 9          6      8
## 10         4      4
## 11         4      6
## 12         6      8
## 13         6      8
## 14         4      6
## 15         5      1
```

```r
#user defined function - if statement checks if the values in the diabetes_binary are equal to 1
#and if they are the function returns "yes" and the else statement returns "no" if the value of
#diabetes_binary is NOT equal to 1
has_diabetes = function(Diabetes_binary) {
  if (Diabetes_binary == 1) {
    return("Yes")
  } else {
    return("No")
  }
}
```

```r
# prints the result of the has_diabetes function at the 1st row in the diabetes_binary column
print(has_diabetes(1))
```

```
## [1] "Yes"
```

```r
# dependent variable is diabetes_binary, the rest are independent variables
# selects the HighBP, HighChol, BMI, and Smoker columns to make the independent variables
diabetes_predictors = Diabetes_binary_data[, c("HighBP", "HighChol", "BMI", "Smoker")]
# selects the Diabetes_diagnosis column to make the dependent variable, named diabetes_diagnosis
diabetes_diagnosis = Diabetes_binary_data$Diabetes_binary
# combines the columns into a new dataframe called diabetes_analysis
```

```r
diabetes_analysis = cbind(diabetes_predictors, diabetes_diagnosis)
head(diabetes_analysis)
```

```
##   HighBP HighChol BMI Smoker diabetes_diagnosis
## 1      1        0  26      0                  0
## 2      1        1  26      1                  0
## 3      0        0  26      0                  0
## 4      1        1  28      1                  0
## 5      0        0  29      1                  0
## 6      0        0  18      0                  0
```

```r
# omits any missing values in the dataset
Diabetes_binary_data = na.omit(Diabetes_binary_data)
nrow(Diabetes_binary_data)
```

```
## [1] 70692
```

```r
# identifies duplicated data in data set and then returns the number of duplicate rows
sum(duplicated(Diabetes_binary_data))
```

```
## [1] 1635
```

```r
# removes any duplicates in data set and returns only the ones that are distinct and displays
# only the first few rows
Diabetes_binary_data = Diabetes_binary_data %>% distinct()
head(Diabetes_binary_data)
```

```
##   Diabetes_binary HighBP HighChol CholCheck BMI Smoker Stroke
## 1               0      1        0         1  26      0      0
## 2               0      1        1         1  26      1      1
## 3               0      0        0         1  26      0      0
## 4               0      1        1         1  28      1      0
## 5               0      0        0         1  29      1      0
## 6               0      0        0         1  18      0      0
##   HeartDiseaseorAttack PhysActivity Fruits Veggies HvyAlcoholConsump
## 1                    0            1      0       1                 0
## 2                    0            0      1       0                 0
## 3                    0            1      1       1                 0
## 4                    0            1      1       1                 0
## 5                    0            1      1       1                 0
## 6                    0            1      1       1                 0
##   AnyHealthcare NoDocbcCost GenHlth MentHlth PhysHlth DiffWalk Sex Age
## 1             1           0       3        5       30        0   1   4
## 2             1           0       3        0        0        0   1  12
## 3             1           0       1        0       10        0   1  13
## 4             1           0       3        0        3        0   1  11
## 5             1           0       2        0        0        0   0   8
## 6             0           0       2        7        0        0   0   1
##   Education Income
## 1         6      8
## 2         6      8
```

```
## 3         6      8
## 4         6      8
## 5         5      8
## 6         4      7
```

```
# arranges data by descending age and displays only the first few rows
Diabetes_binary_data = Diabetes_binary_data %>% arrange(desc(Age))
head(Diabetes_binary_data)
```

```
##   Diabetes_binary HighBP HighChol CholCheck BMI Smoker Stroke
## 1               0      0        0         1  26      0      0
## 2               0      0        1         1  26      1      0
## 3               0      1        0         1  22      0      0
## 4               0      0        0         1  26      0      0
## 5               0      0        0         1  22      0      0
## 6               0      1        0         1  25      0      0
##   HeartDiseaseorAttack PhysActivity Fruits Veggies HvyAlcoholConsump
## 1                    0            1      1       1                 0
## 2                    0            1      1       1                 1
## 3                    0            1      1       1                 0
## 4                    0            1      1       1                 0
## 5                    0            1      1       1                 0
## 6                    0            0      0       0                 0
##   AnyHealthcare NoDocbcCost GenHlth MentHlth PhysHlth DiffWalk Sex Age
## 1             1           0       1        0       10        0   1  13
## 2             1           0       1        0        0        0   1  13
## 3             1           0       3        0        0        1   1  13
## 4             1           0       3        0        0        0   0  13
## 5             1           0       1        0        0        0   0  13
## 6             1           0       4       14       14        1   0  13
##   Education Income
## 1         6      8
## 2         5      6
## 3         6      6
## 4         4      3
## 5         6      4
## 6         5      7
```

```
# renames columns
names(Diabetes_binary_data)[names(Diabetes_binary_data) == "Diabetes_binary"] = "Diabetes_diagnosis"
names(Diabetes_binary_data)[names(Diabetes_binary_data) == "Veggies"] = "Eats_enough_veggies"
names(Diabetes_binary_data)[names(Diabetes_binary_data) == "GenHlth"] = "General_health"
names(Diabetes_binary_data)[names(Diabetes_binary_data) == "AnyHealthcare"] = "Has_any_healthcare"
names(Diabetes_binary_data)[names(Diabetes_binary_data) == "MentHlth"] = "Mental_health"
```

```
# make a new column called socioeconomic_score and it will be the result of income x education
# and displays first few rows
Diabetes_binary_data = Diabetes_binary_data %>% mutate(SocioeconomicScore = Income*Education)
head(Diabetes_binary_data)
```

```
##   Diabetes_diagnosis HighBP HighChol CholCheck BMI Smoker Stroke
## 1                  0      0        0         1  26      0      0
```

```
## 2                  0      0        1         1  26      1      0
## 3                  0      1        0         1  22      0      0
## 4                  0      0        0         1  26      0      0
## 5                  0      0        0         1  22      0      0
## 6                  0      1        0         1  25      0      0
##   HeartDiseaseorAttack PhysActivity Fruits Eats_enough_veggies
## 1                    0            1      1                   1
## 2                    0            1      1                   1
## 3                    0            1      1                   1
## 4                    0            1      1                   1
## 5                    0            1      1                   1
## 6                    0            0      0                   0
##   HvyAlcoholConsump Has_any_healthcare NoDocbcCost General_health Mental_health
## 1                 0                  1           0              1             0
## 2                 1                  1           0              1             0
## 3                 0                  1           0              3             0
## 4                 0                  1           0              3             0
## 5                 0                  1           0              1             0
## 6                 0                  1           0              4            14
##   PhysHlth DiffWalk Sex Age Education Income SocioeconomicScore
## 1       10        0   1  13         6      8                48
## 2        0        0   1  13         5      6                30
## 3        0        1   1  13         6      6                36
## 4        0        0   0  13         4      3                12
## 5        0        0   0  13         6      4                24
## 6       14        1   0  13         5      7                35
```

```
# create a training set using random number generator engine
set.seed(1234)
training_set = Diabetes_binary_data %>% sample_frac(0.80)
nrow(training_set)
```

```
## [1] 55246
```

```
# Prints the summary statistics of data set
view_all = summary(Diabetes_binary_data)
print(view_all)
```

```
##  Diabetes_diagnosis     HighBP          HighChol        CholCheck
##  Min.   :0.0000     Min.   :0.0000   Min.   :0.0000   Min.   :0.0000
##  1st Qu.:0.0000     1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:1.0000
##  Median :1.0000     Median :1.0000   Median :1.0000   Median :1.0000
##  Mean   :0.5082     Mean   :0.5712   Mean   :0.5313   Mean   :0.9748
##  3rd Qu.:1.0000     3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:1.0000
##  Max.   :1.0000     Max.   :1.0000   Max.   :1.0000   Max.   :1.0000
##       BMI            Smoker           Stroke        HeartDiseaseorAttack
##  Min.   :12.00   Min.   :0.0000   Min.   :0.00000   Min.   :0.0000
##  1st Qu.:25.00   1st Qu.:0.0000   1st Qu.:0.00000   1st Qu.:0.0000
##  Median :29.00   Median :0.0000   Median :0.00000   Median :0.0000
##  Mean   :29.96   Mean   :0.4819   Mean   :0.06364   Mean   :0.1509
##  3rd Qu.:33.00   3rd Qu.:1.0000   3rd Qu.:0.00000   3rd Qu.:0.0000
##  Max.   :98.00   Max.   :1.0000   Max.   :1.00000   Max.   :1.0000
##   PhysActivity        Fruits       Eats_enough_veggies HvyAlcoholConsump
```

```
##  Min.   :0.0000   Min.   :0.0000   Min.   :0.0000      Min.   :0.00000
##  1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:1.0000      1st Qu.:0.00000
##  Median :1.0000   Median :1.0000   Median :1.0000      Median :0.00000
##  Mean   :0.6965   Mean   :0.6057   Mean   :0.7841      Mean   :0.04351
##  3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:1.0000      3rd Qu.:0.00000
##  Max.   :1.0000   Max.   :1.0000   Max.   :1.0000      Max.   :1.00000
##  Has_any_healthcare  NoDocbcCost      General_health  Mental_health
##  Min.   :0.0000     Min.   :0.00000   Min.   :1.000   Min.   : 0.00
##  1st Qu.:1.0000     1st Qu.:0.00000   1st Qu.:2.000   1st Qu.: 0.00
##  Median :1.0000     Median :0.00000   Median :3.000   Median : 0.00
##  Mean   :0.9539     Mean   :0.09614   Mean   :2.864   Mean   : 3.84
##  3rd Qu.:1.0000     3rd Qu.:0.00000   3rd Qu.:4.000   3rd Qu.: 3.00
##  Max.   :1.0000     Max.   :1.00000   Max.   :5.000   Max.   :30.00
##     PhysHlth         DiffWalk           Sex              Age
##  Min.   : 0.000   Min.   :0.0000   Min.   :0.0000   Min.   : 1.000
##  1st Qu.: 0.000   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.: 7.000
##  Median : 0.000   Median :0.0000   Median :0.0000   Median : 9.000
##  Mean   : 5.945   Mean   :0.2586   Mean   :0.4565   Mean   : 8.604
##  3rd Qu.: 6.000   3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:11.000
##  Max.   :30.000   Max.   :1.0000   Max.   :1.0000   Max.   :13.000
##    Education       Income      SocioeconomicScore
##  Min.   :1.0   Min.   :1.000   Min.   : 1.0
##  1st Qu.:4.0   1st Qu.:4.000   1st Qu.:16.0
##  Median :5.0   Median :6.000   Median :30.0
##  Mean   :4.9   Mean   :5.651   Mean   :28.7
##  3rd Qu.:6.0   3rd Qu.:8.000   3rd Qu.:40.0
##  Max.   :6.0   Max.   :8.000   Max.   :48.0
```

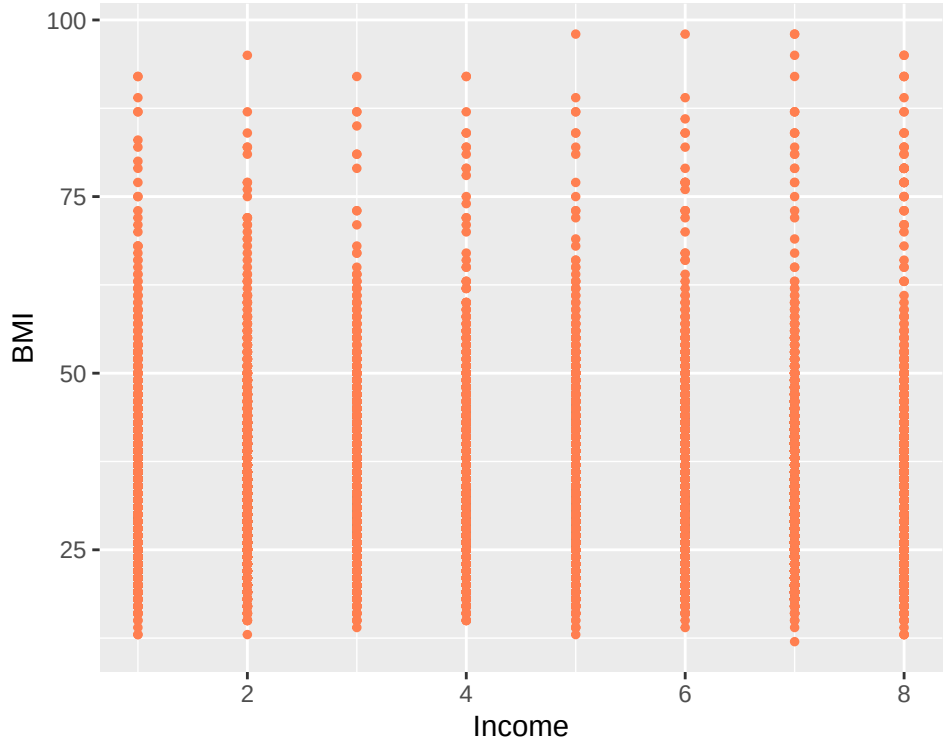
```r
# Finds the mean of the BMI column
BMI_mean = mean(Diabetes_binary_data$BMI)
```

```r
# Finds the median for the BMI column
BMI_median = median(Diabetes_binary_data$BMI)
```

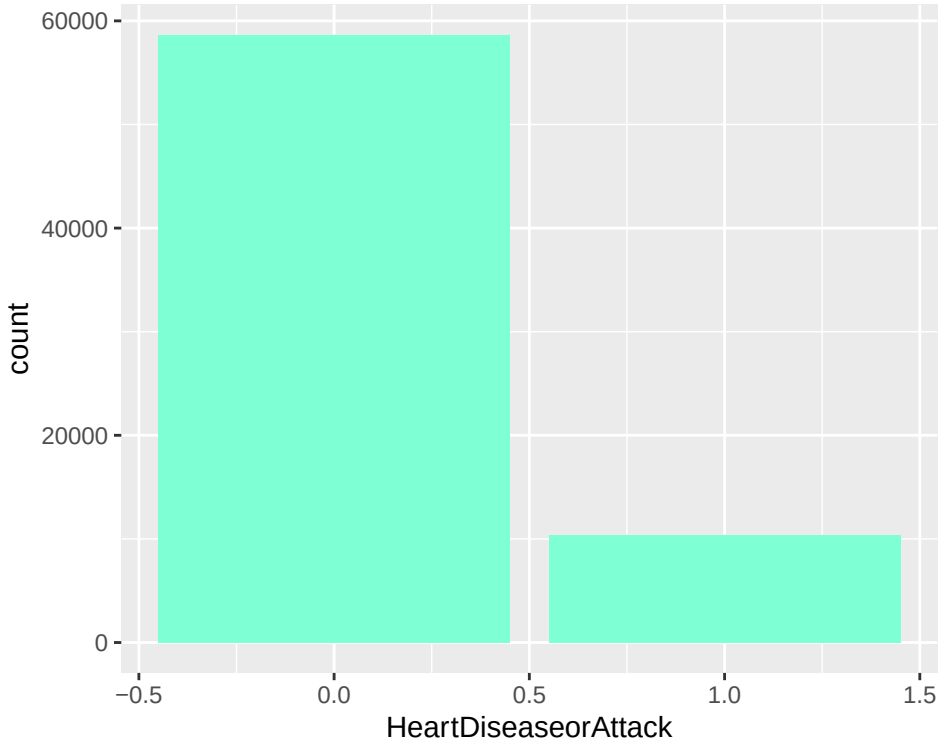
```r
# Finds mode of the BMI column using a user-defined function
# unique(x) extracts the unique values in the BMI column
# match(x, values) matches each observation to its position in the list of distinct
# values, where values = unique(x)
# tabulate() counts how many times each distinct value occurs
# which.max() identifies the position of the highest frequency
# values[] returns the value at that position in the newly created vector
find_mode = function(x) {
  values = unique(x)
  values[which.max(tabulate(match(x, values)))]
}
# calls the find_mode function passing the BMI data from the data set and stores it in
# a variable called "BMI_mode"
BMI_mode = find_mode(Diabetes_binary_data$BMI)
```

```r
# Finds range of BMI column data
BMI_range = range(Diabetes_binary_data$BMI)
```

```
# Creates a scatter plot for the income and BMI column data
ggplot(Diabetes_binary_data, aes(x = Income, y = BMI)) + geom_point(size = 1, color = "coral")
```




```
# Creates a bar graph for the HeartDiseaseorAttack column data
ggplot(Diabetes_binary_data, aes(x = HeartDiseaseorAttack)) + geom_bar(fill = "aquamarine")
```




```
# Finds the pearson correlation between Diabetes_diagnosis and BMI
BMI_corr = cor(Diabetes_binary_data$Diabetes_diagnosis, Diabetes_binary_data$BMI, method = "pearson")
```